Spherical coordinate system

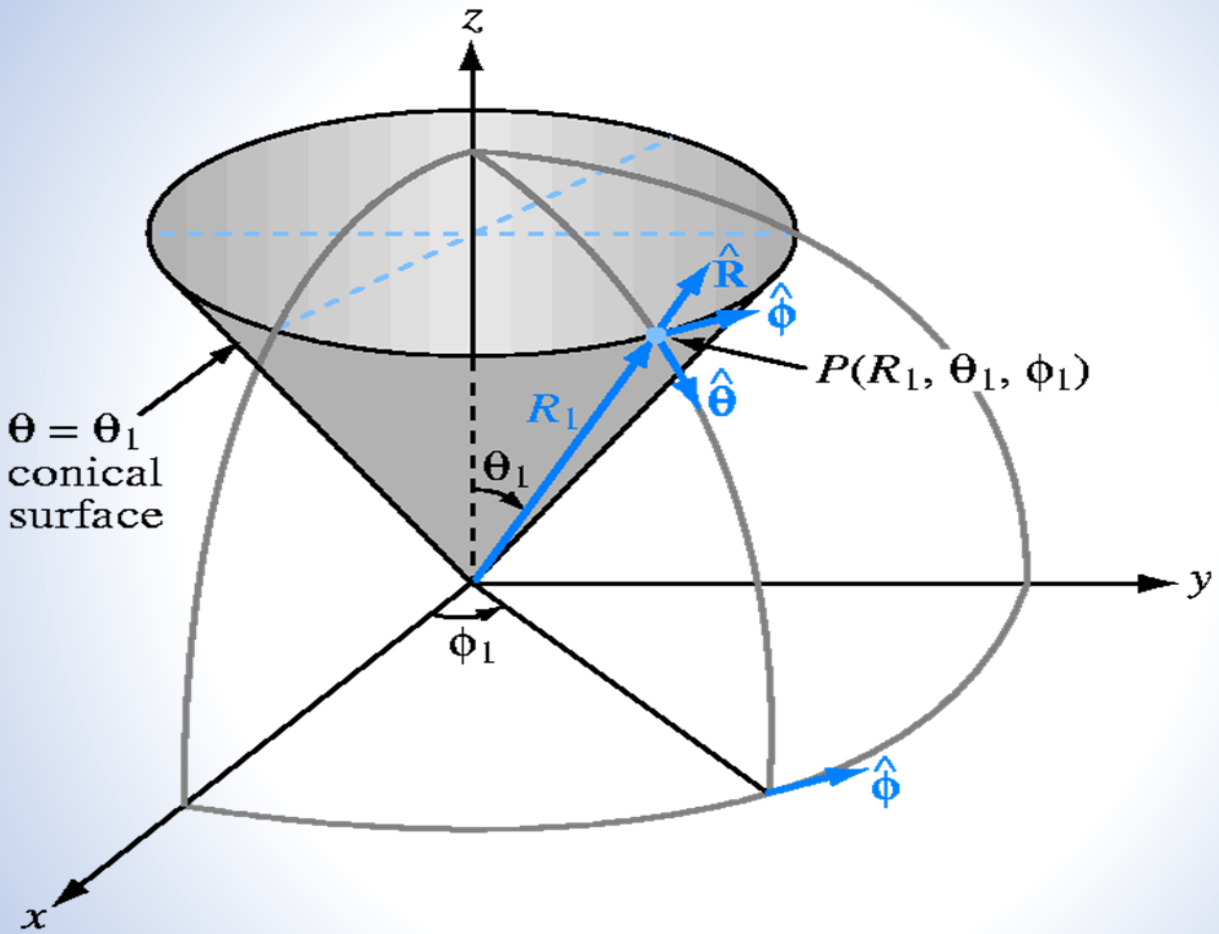
Dr. Sampath Kumar

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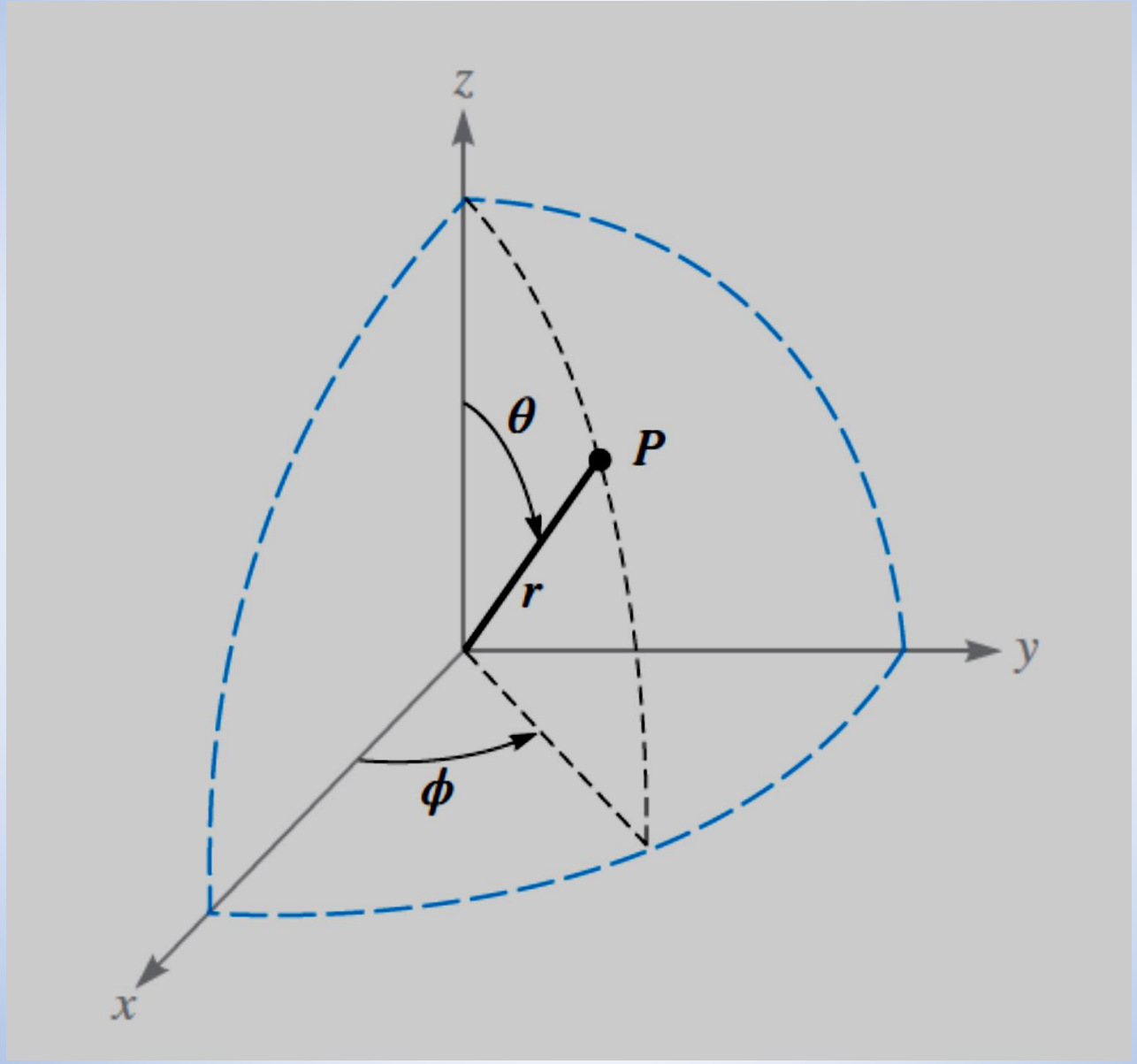
MIT, Manipal

The spherical coordinate system

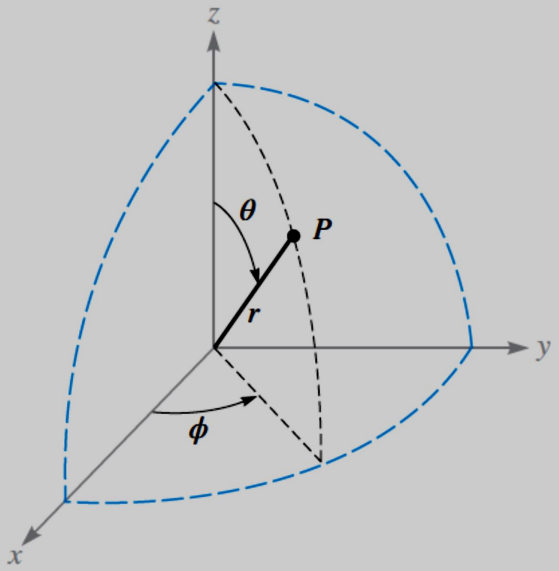


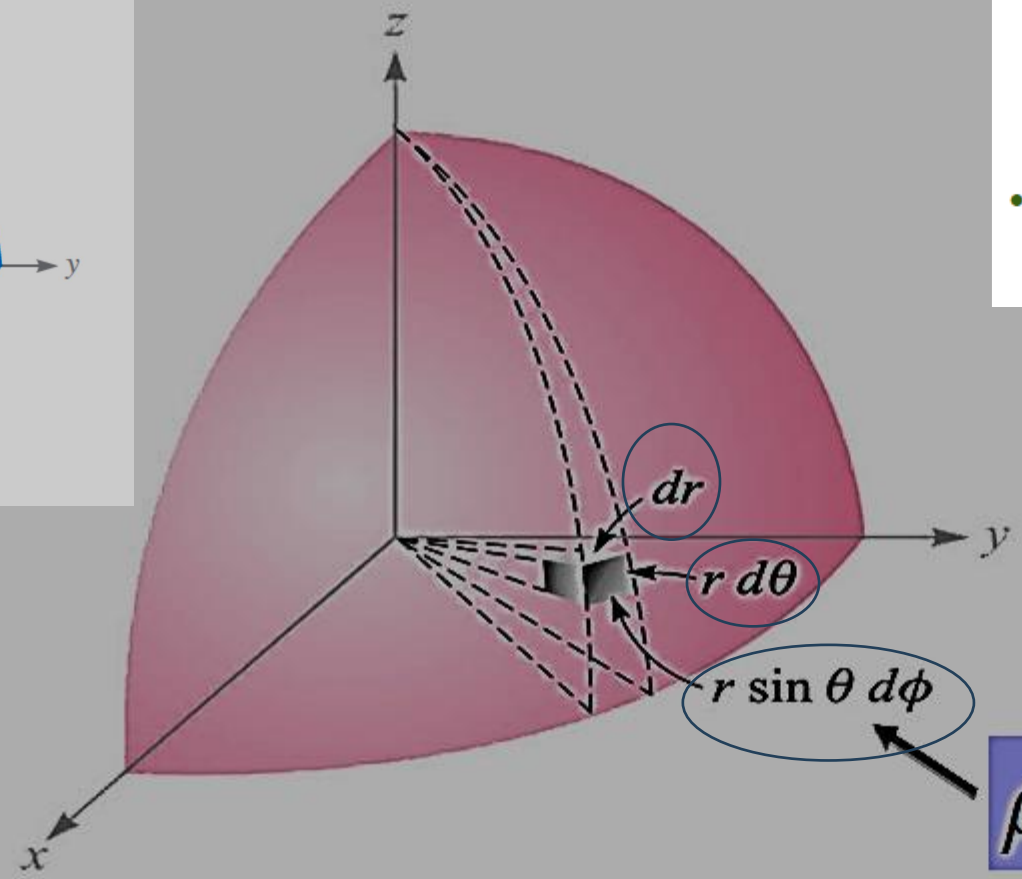
$$0 \leq r < \infty$$
$$0 \leq \theta \leq \pi$$
$$0 \leq \phi < 2\pi$$

https://mathinsight.org/applet/spherical_coordinates_constant_rho
https://mathinsight.org/applet/spherical_coordinates_constant_theta
https://mathinsight.org/applet/spherical_coordinates_constant_phi


z
θ
P
r
y
ϕ
x

Differential elements



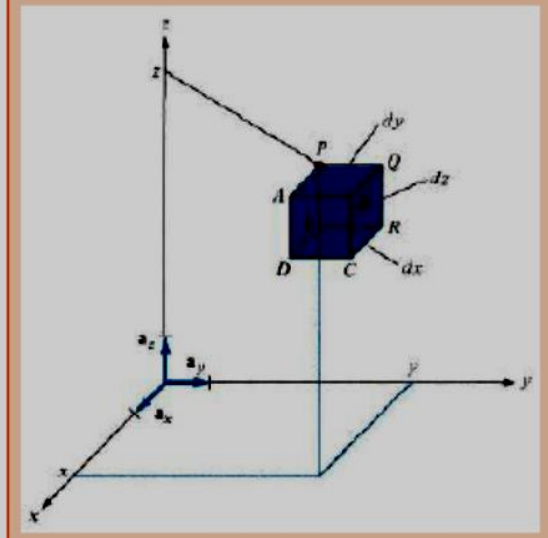
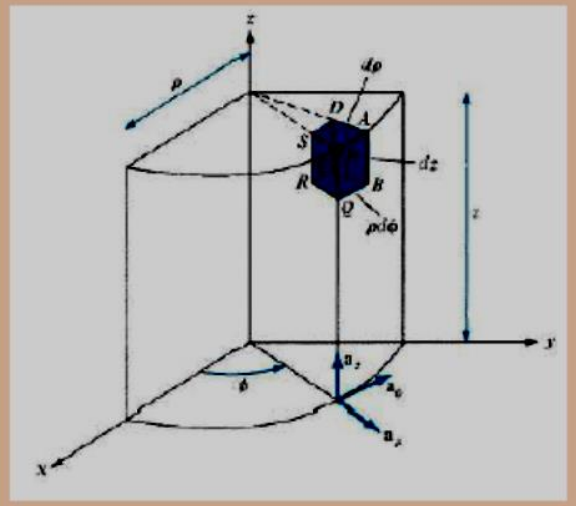
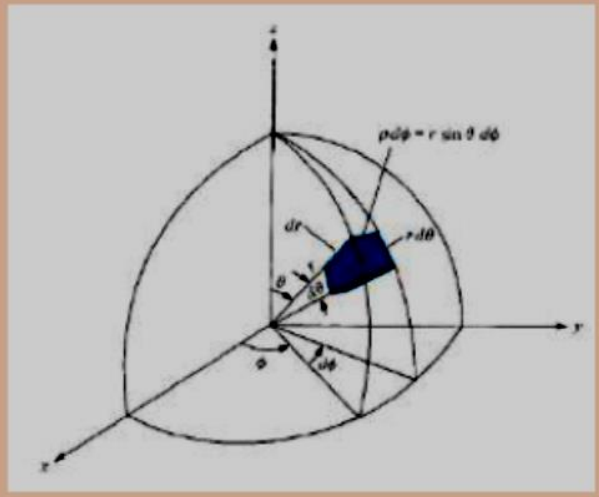


- **Differential surface units:**
  $dr \cdot rd\theta$
  $dr \cdot r\sin\theta d\phi$
  $rd\theta \cdot r\sin\theta d\phi$
- **Differential volume unit :**
  $dr \cdot rd\theta \cdot r\sin\theta d\phi$

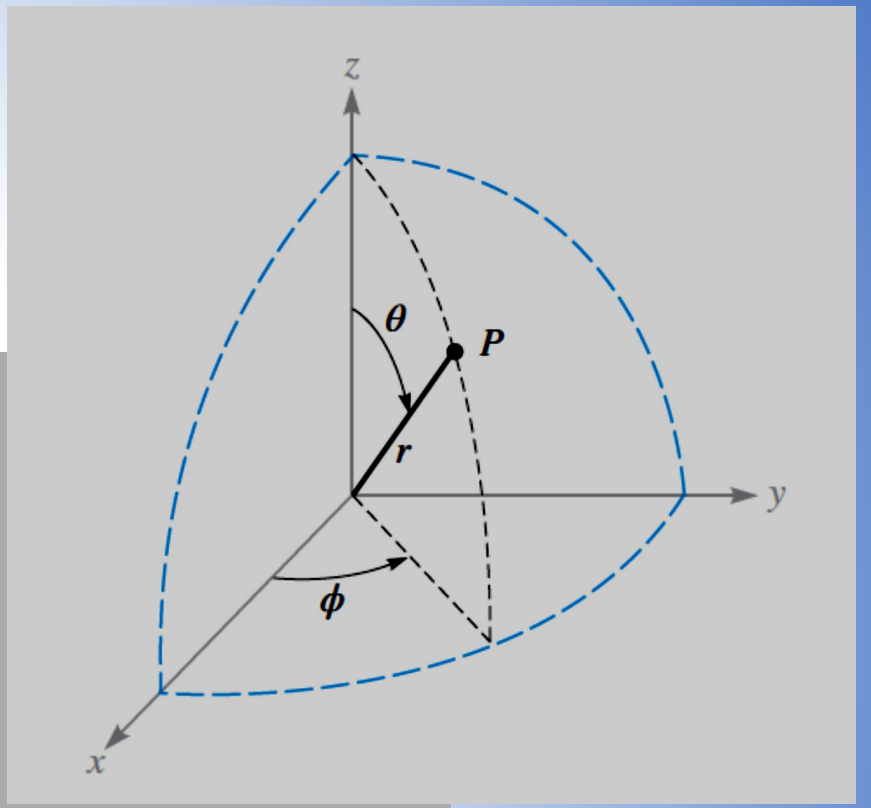
$$\rho d\phi = r\sin\theta d\phi$$

Differential displacement : $\vec{dl} = dra_r + rd\theta a_\theta + r\sin\theta d\phi a_\phi$

Differential length, surface and volume for Cartesian, Cylindrical and Spherical Coordinate Systems

Cartesian	Cylindrical	Spherical
		
$d\boldsymbol{l} = dx\,\boldsymbol{a_x} + dy\,\boldsymbol{a_y} + dz\,\boldsymbol{a_z}$	$d\boldsymbol{l} = d\rho\,\boldsymbol{a_\rho} + \rho\,d\emptyset\,\boldsymbol{a_\emptyset} + dz\,\boldsymbol{a_z}$	$d\boldsymbol{l} = dr\,\boldsymbol{a_r} + r\,d\theta\,\boldsymbol{a_\theta} + r\sin\theta\;d\emptyset\,\boldsymbol{a_\emptyset}$
$d\boldsymbol{S} = dy\,dz\,\boldsymbol{a_x} = dx\,dz\,\boldsymbol{a_y} = dx\,dy\,\boldsymbol{a_z}$	$d\boldsymbol{S} = \rho\,d\emptyset\,dz\,\boldsymbol{a_\rho} = d\rho\,dz\,\boldsymbol{a_\emptyset} = \rho\,d\rho\,d\emptyset\,\boldsymbol{a_z}$	$d\boldsymbol{S} = r^2\sin\theta\,d\theta\,d\emptyset\,\boldsymbol{a_r} = r\sin\theta\;dr\,d\emptyset\,\boldsymbol{a_\theta} = r\,dr\,d\theta\,\boldsymbol{a_\emptyset}$
$dv = dx\,dy\,dz$	$dv = \rho\,d\rho\,d\emptyset\,dz$	$dv = r^2\sin\theta\;dr\,d\theta\,d\emptyset$

The Spherical Coordinate System



- **Relation between the rectangular and the spherical coordinate systems**

$x = r\sin\theta\cos\phi$	$r = \sqrt{x^2 + y^2 + z^2}, \quad r \geq 0$
$y = r\sin\theta\sin\phi$	$\theta = \cos^{-1}\dfrac{z}{\sqrt{x^2 + y^2 + z^2}}, \quad 0° \leq \theta \leq 180°$
$z = r\cos\theta$	$\phi = \tan^{-1}\dfrac{y}{x}$

D1.7. Given the two points, $C(-3, 2, 1)$ and $D(r = 5, \theta = 20°, \phi = -70°)$, find: (*a*) the spherical coordinates of C; (*b*) the rectangular coordinates of D; (*c*) the distance from C to D.

a)

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{9 + 4 + 1} = 3.74$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{1}{3.74}\right) = 74.5°$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-2}{3}\right) = 146.3°$$

b)

$$x = r\sin\theta\cos\phi = 5\sin 20°\cos(-70°) = 0.585$$

$$y = r\sin\theta\sin\phi = 5\sin 20°\sin(-70°) = -1.607$$

$$z = r\cos\theta = 5\cos 20° = 4.7$$

c)

$$R_{CD} = C_{(-3,2,1)} - D_{(0.585, -1.607, 4.7)}$$

$$= 6.29$$

Table 1.2 Dot products of unit vectors in spherical and rectangular coordinate systems

	$\mathbf{a}_r$	$\mathbf{a}_\theta$	$\mathbf{a}_\phi$
$\mathbf{a}_x \cdot$	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$
$\mathbf{a}_y \cdot$	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$
$\mathbf{a}_z \cdot$	$\cos\theta$	$-\sin\theta$	0

D1.8. Transform the following vectors to spherical coordinates at the points given: (*a*) $10\mathbf{a}_x$ at $P(x=-3, y=2, z=4)$; (*b*) $10\mathbf{a}_y$ at $Q(\rho=5, \phi=30°, z=4)$; (*c*) $10\mathbf{a}_z$ at $M(r=4, \theta=110°, \phi=120°)$.

a) $\vec{A} = 10\hat{a}_x = A_r\hat{a}_r + A_\theta\hat{a}_\theta + A_\phi\hat{a}_\phi$

$A_r = \vec{A}\cdot\hat{a}_r = 10\hat{a}_x\cdot\hat{a}_r = 10\sin\theta\cos\phi$

$A_\theta = \vec{A}\cdot\hat{a}_\theta = 10\hat{a}_x\cdot\hat{a}_\theta = 10\cos\theta\cos\phi$

$A_\phi = \vec{A}\cdot\hat{a}_\phi = 10\hat{a}_x\cdot\hat{a}_\phi = -10\sin\phi$

Table 1.2 Dot products of unit vectors in spherical and rectangular coordinate systems

	$\mathbf{a}_r$	$\mathbf{a}_\theta$	$\mathbf{a}_\phi$
$\mathbf{a}_x\cdot$	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$
$\mathbf{a}_y\cdot$	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$
$\mathbf{a}_z\cdot$	$\cos\theta$	$-\sin\theta$	0

Substitute the values of θ & ϕ in A_r, A_θ, A_ϕ

$r = \sqrt{9+4+16} = 5.385$

$\theta = \cos^{-1}\left(\frac{4}{5.385}\right) = 42.03°$

$\phi = \tan^{-1}\left(\frac{2}{-3}\right) = -33.7°$

$\phi = 146.3°$

We get $\vec{A} = -5.57\hat{a}_r - 6.18\hat{a}_\theta - 5.55\hat{a}_\phi$

D1.8. Transform the following vectors to spherical coordinates at the points given: (*a*) $10\mathbf{a}_x$ at $P(x=-3, y=2, z=4)$; (*b*) $10\mathbf{a}_y$ at $Q(\rho=5, \phi=30°, z=4)$; (*c*) $10\mathbf{a}_z$ at $M(r=4, \theta=110°, \phi=120°)$.

- {Part (b) – try; Ans: $3.90\mathbf{a}_r + 3.12\mathbf{a}_\theta + 8.66\mathbf{a}_\phi$}
- Part (c):

$$\vec{A} = 10\hat{a}_z = A_r\hat{a}_r + a_\theta\hat{a}_\theta + A_\phi\hat{a}_\phi$$

$$A_r = \vec{A}\cdot\hat{a}_r = 10\hat{a}_z\cdot\hat{a}_r = 10\cos\theta$$

$$A_\theta = \vec{A}\cdot\hat{a}_\theta = 10\hat{a}_z\cdot\hat{a}_\theta = -10\sin\theta$$

$$A_\phi = \vec{A}\cdot\hat{a}_\phi = 10\hat{a}_z\cdot\hat{a}_\phi = 0$$

Table 1.2 Dot products of unit vectors in spherical and rectangular coordinate systems

	$\mathbf{a}_r$	$\mathbf{a}_\theta$	$\mathbf{a}_\phi$
$\mathbf{a}_x\cdot$	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$
$\mathbf{a}_y\cdot$	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$
$\mathbf{a}_z\cdot$	$\cos\theta$	$-\sin\theta$	0

Substitute value of $\theta = 110°$

we get $\vec{A} = -3.42\hat{a}_r - 9.4\hat{a}_\theta$


Thank
you